EC 704 - Macroeconomic Theory 2

Problem Set 2

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Problem 1

Note that each period is independent, and therefore separation decisions only depends z in the period. Let separation decision be represented by $s(z) \in \{0, 1\}$.

Part 1

Under given condition

$$s(z) = \begin{cases} 0 & \text{if } z \geq \hat{z} \\ 1 & \text{if } z < \hat{z} \end{cases}$$

The value of a job for the firm is

$$J(z) = z - w(z) + \beta \mathbb{E}\left[\max\{J(z'), V - \xi\}\right]$$

The value of vacancy is

$$V = c + \beta \mathbb{E}[q(\theta) \max\{J(z'), V - \xi\} + (1 - q(\theta))V]$$

The value of an employed worker is

$$E(z) = w(z) + \beta \mathbb{E}[\max\{E(z'), U\}]$$

The value of an unemployed worker is

$$U = b + \beta \mathbb{E}[(1 - f(\theta))U + f(\theta) \max\{E(z'), U\}]$$

Part 2

Due to free entry, $V = 0$. We have

$$\begin{aligned} S(z) &= E(z) + J(z) - U - (V - \xi) \\ &= z - b + (1 - \beta)\xi + \beta \mathbb{E}\left[\max\{J(z') + \xi, 0\} + (1 - f(\theta)) \max\{E(z') - U, 0\}\right] \\ &= z - b + (1 - \beta)\xi + \beta \mathbb{E}\left[\max\{J(z') + \xi, 0\} + \max\{E(z') - U, 0\} - f(\theta) \max\{E(z') - U, 0\}\right] \end{aligned}$$

Note that $s(z) = 0 \iff J(z') + \xi \geq 0, E(z') - U \geq 0$.
Therefore, $\max\{J(z') + \xi, 0\} + \max\{E(z') - U, 0\} = \max\{J(z') + \xi + E(z') - U, 0\} = \max\{S(z'), 0\}$.
Hence,

$$\begin{aligned} S(z) &= z - b + (1 - \beta)\xi + \beta \mathbb{E}\left[\max\{S(z'), 0\} - f(\theta) \max\{E(z') - U, 0\}\right] \\ &= z - b + (1 - \beta)\xi + \beta \int_{\hat{z}}^{\infty} S(\tilde{z}) dG(\tilde{z}) - \beta f(\theta) \int_{\hat{z}}^{\infty} (E(\tilde{z}) - U) dG(\tilde{z}) \end{aligned} \tag{1-1}$$

Part 3

For any $z > z'$, the firm and worker accepts the match and generate surplus $S(z)$. From the expression above, $S'(z) = 1$. For $z = z'$, the firm and worker are indifferent between separating and working. Therefore, the surplus must be zero. Otherwise, is surplus was positive, they would accept the match and split the surplus. Hence

$$S(z) = z - \hat{z}$$

Part 4

Nash bargaining problem

$$\begin{aligned} w(z) &= \text{argmax}_{\{E(z)-U, J(z)+\xi\}} (E(z) - U)^{\gamma} (J(z) + \xi)^{1-\gamma} \\ s.t. \quad & S(z) = z - \hat{z} = E(z) - U + J(z) + \xi \end{aligned}$$

which is equivalent to

$$\begin{aligned} w(z) &= \text{argmax}_{\{E(z)-U, J(z)+\xi\}} \gamma \log(E(z) - U) + (1-\gamma) \log(J(z) + \xi) \\ s.t. \quad & S(z) = z - \hat{z} = E(z) - U + J(z) + \xi \end{aligned}$$

Using lagrnage multiplies λ, the first order conditions are given by

$$\begin{aligned} \gamma \frac{1}{E(z) - U} &= \lambda \\ (1-\gamma) \frac{1}{J(z) + \xi} &= \lambda \end{aligned}$$

Using $S(z) = E(z) - U + J(z) + \xi = \frac{1}{\lambda}$, we have

$$\begin{aligned} E(z) - U &= \gamma S(z) \\ J(z) + \xi &= (1-\gamma) S(z) \end{aligned}$$

Part 5

Substituting $E(z) - U = \gamma S(z)$ in equation 1-1, and setting $S(\hat{z}) = 0$, we get

$$0 = \hat{z} - b + (1-\beta)\xi + \beta(1 - \gamma f(\theta)) \int_{\hat{z}}^{\infty} (\tilde{z} - \hat{z}) dG(\tilde{z}) \tag{1-6}$$

we can rewrite this as

$$f(\theta) = \frac{\hat{z} - b + (1-\beta)\xi}{\gamma\beta \int_{\hat{z}}^{\infty} (\tilde{z} - \hat{z}) dG(\tilde{z})} + \frac{1}{\gamma}$$

If $\hat{z}$ increases, then the integral value decreases and therefore $f(\theta)$ increases. This gives a positive relationship. An increase in vacancies or decrease in unemployment increases the job finding rate for workers, which results in increase in outside options for workers. Due to higher outside option, workers demand a higher wage that increases the cut-off productivity $\hat{z}$.

Part 6

As described in part 1, we have the value of vacancy

$$V = c + \beta\mathbb{E}[q(\theta)\max\{J(z'), V - \xi\} + (1 - q(\theta))V]$$

Setting $V = 0$, this gives

$$\begin{aligned} c &= \beta q(\theta)\mathbb{E}[\max\{J(z') + \xi, 0\} - \xi] \\ &= \beta q(\theta)\mathbb{E}[(1-\gamma)\max\{S(z')\} - \xi] \\ &= \beta q(\theta)\left((1-\gamma)\int_{\hat{z}}^{\infty}(\tilde{z} - \hat{z})dG(\tilde{z}) - \xi\right) \end{aligned} \tag{1-7}$$

An increase in $\hat{z}$ decreases the intgral, and therefore requires a higher value of $q(\theta)$. Since q is a decreasing function of θ, this defines a decreasing relationship between θ and $\hat{z}$. An increase in vacancies or decrease in unemployment (tighter labor market) results in a lower worker finding rate of firms, and therefore firms are willing to accept matches with lower productivity.

Part 7

Equation (1-6) implies a increasing relationship between $\hat{z}$ and θ due to workers demand for higher surplus when markets become tighter. Equation (1-7) shows a decreasing relationship between $\hat{z}$ and θ due to firms accepting a lower productivity workers as markets become tighter. Since both these equations hold in equilibrium, the unique solution is given by the intersection.

Part 8

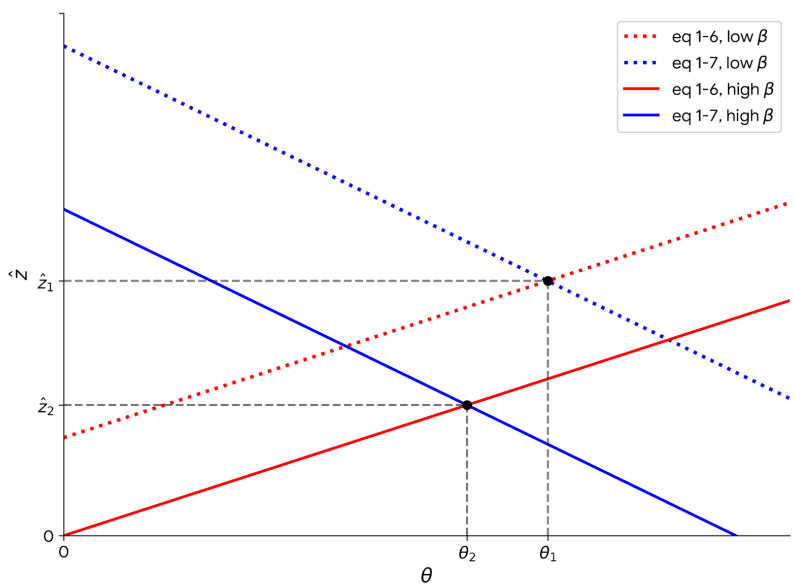

Figure 1: Effects of increasing ξ on $\hat{z}$ and θ

From above figure, a larger ξ leads to a decrease in cut-off productivity $\hat{z}$. The effects on θ are ambiguous. Due to higher job firing cost, the number of vacancies decrease as firms are less willing

to fire workers and are more flexible with hiring workers. But at the same time, number of firms decreases since firms future profits of hiring worker now decrease. The number of jobs decrease but the firms become more flexible in hiring. Therefore, whether labor markets become tigher or looser depends on the relative importance of these two effects, which depends on parameters in equations (1-6) and (1-7) and the productivity distribution G.

In each period, the productivity is realized independently of prior periods. An employed workers in period t become unemployed in period $t+1$ with probability $\mathbb{P}(z < \hat{z})$. An unemployed worker in period t received a job offer with probability $f(\theta_t)$, and accepts it with probability $P(z \geq \hat{z})$. Therefore, in this model, the endogenous separation rate is $s = \mathbb{P}(z < \hat{z})$ and the endogenous job finding rate is $\tilde{f} = f(\theta)\mathbb{P}(z \geq \hat{z})$. The steady state unemployment is given by

$$\frac{u}{1-u} = \frac{s}{\tilde{f}} = \frac{\mathbb{P}(z < \hat{z})}{f(\theta)\left(1 - \mathbb{P}(z < \hat{z})\right)}$$

When ξ increases, $\hat{z}$ decreases and $\frac{\mathbb{P}(z<\hat{z})}{(1-\mathbb{P}(z<\hat{z}))}$ decreases. However, change is $f(\theta)$ depends on the distribution G through equations (1-6) and (1-7), and therefore unemployment change is also dependent of distribution G.

Problem 2

Part 1

Let us denote the unemployment shares of g type workers by u_g. The total unemployment is the economu is given by $u = 0.5u_1 + 0.5u_2$. The market tightness is $\theta = \frac{v}{u}$.

In the McGall + DMP model with homogenous workers, the wage is given by $w(z) = \gamma z + (1-\gamma)rU$. With heterogenous workers, since U_g is different for differnet worker types, the wage would vary according to worker type. Intuitively, if firms match with worker with more outside option, firms would be villing to pay more to avoid the risk of bearing vacancy cost.

The flow value of a filled job with worker of tpye g is

$$(r+s)J_g(z) = z - w_g(z) + sV \tag{1}$$

The flow value of a vacancy is

$$rV = -c + Aq(\theta)\left[\frac{u_1}{u_1+u_2}\int \max\{J_1(z) - V, 0\}dG(z) + \frac{u_2}{u_1+u_2}\int \max\{J_2(z) - V, 0\}dG(z)\right] \tag{2}$$

The flow value of unemployed worker of type g is

$$rU_g = b_g + Af(\theta)\int \max\{E_g(z) - U_g, 0\}dG(z) \tag{3}$$

The flow value of employed worker of type g is

$$(r+s)E_g(z) = w_g(z) + sU_g \tag{4}$$

Free entry conditions imply that

$$V = 0$$

The surplus of match between a firm and type g worker is

$$S_g(z) = \frac{z}{r+s} - \frac{r}{r+s}U_g \tag{5}$$

The reservation match quality for type g worker satisfies $S(z_g^R) = 0$, which gives

$$z_g^R = rU_g$$

We have the following steady state equilibrium conditions

$$z_1^R - b_1 = \frac{\gamma Af(\theta)}{r+s}\int_{z_1^R}(z - z_1^R)dG(z) \tag{6}$$

$$z_2^R - b_2 = \frac{\gamma Af(\theta)}{r+s}\int_{z_2^R}(z - z_2^R)dG(z) \tag{7}$$

$$c = \frac{(1-\gamma)Aq(\theta)}{r+s}\left[\frac{u_1}{u_1+u_2}\int_{z_1^R}(z - z_1^R)dG(z) + \frac{u_2}{u_1+u_2}\int_{z_2^R}(z - z_2^R)dG(z)\right] \tag{8}$$

Additionally, the steady state employment levels are given by

$$\frac{1-u_1}{u_1} = \frac{Af(\theta)(1-G(z_1^R))}{s} \tag{9}$$

$$\frac{1-u_2}{u_2} = \frac{Af(\theta)(1-G(z_2^R))}{s} \tag{10}$$

$$\tag{11}$$

Part 2

We have the condition $u = 0.5u_1 + 0.5u_2 = 0.05$. Since $b_1 = b_2$ in the given steady state, we have $u_1 = u_2 = 0.05$. This gives

$$f(\theta) = \frac{19s}{1-G(z_1^R)} \tag{12}$$

With $\eta = 0.5$, we also have $q(\theta) = \frac{1}{f(\theta)}$. Therefore, we have the following two equations

$$(r+s)(z_1^R - b_1)(1-G(z_1^R)) = 19\gamma s \int_{z_1^R} (z - z_1^R) dG(z)$$

$$19cs(r+s) = (1-\gamma)(1-G(z_2^R)) \int_{z_2^R} (z - z_2^R) dG(z)$$

Solving these numerically gives

$$\gamma = 0.129122$$

The UE transition rates are given by

$$\theta = 4.866649$$

$$z_1^R = z_1^R = 0.142156$$

$$UE_{10} = UE_{20} = Af(\theta)(1-G(z_1^R)) = s\frac{1-u_1}{u_1} = 0.38$$

Part 3

Using the above computed γ and $\beta_1 = 1.5 * 0.07, \beta_2 = 0.07$, we can compute $\theta, u_1, u_2, z_1^R, z_2^R$ using equations 9, 10, 8, **??** and **??**.

$$u = 0.064857 \tag{13}$$

$$u_1 = 0.078724 \tag{14}$$

$$u_2 = 0.050991 \tag{15}$$

$$z_1^R = 0.152757 \tag{16}$$

$$z_2^R = 0.139221 \tag{17}$$

$$\theta = 3.790107 \tag{18}$$

These gives the UE transition rates

$$UE_{11} = 0.234053 \tag{19}$$
$$UE_{21} = 0.372227 \tag{20}$$
$$\tag{21}$$

Type 1 workers have higher outside options, and are therefore more pickier. They reject more offers and have higher wages. Therefore, type 1 workers have higher unemployment and lower UE transitions that type 2 workers. Unemploymnet rates of type 2 workers is slightly larger than 0.05 since type 1 workers pull down market tightness and therefore decreases matching probability of type 1 workers.

Part 4

The OLS regressions gives $\hat{\beta} = -1.17824$ with $se = 0.03398$ and p-value of $p = 0.0008$.

$\hat{\beta}$ is the combined **Macro+Micro** estimate of elasticity of UE transitions w.r.t outside options.

Part 5

The OLS regressions gives $\check{\beta} = -1.19523$ with $se = 0.05097$ and p-value of $p = 0.0271$. Also $\alpha_1 - \alpha_2 = 0.01033$.

$\check{\beta}$ is the **Micro** estimate of elasticity of UE transitions w.r.t outside options. It is slightly smaller than but not significantly differnet from $\hat{\beta}$. This means that the micro effects dominate over macro effects, and the general equilibrium macro effects of increased outside option (less tighter labor markets and lower match rates) counter the negative effect of outside option on UE transitions.

Part 6

The computed values of $\hat{\beta}$ and $\check{\beta}$ are large, implying that 1% increase in unemployment benefits lead to $\approx 1.17 - 1.19\%$ decrease in unemployment. This is inconsistent with empiricial results.

Problem 3